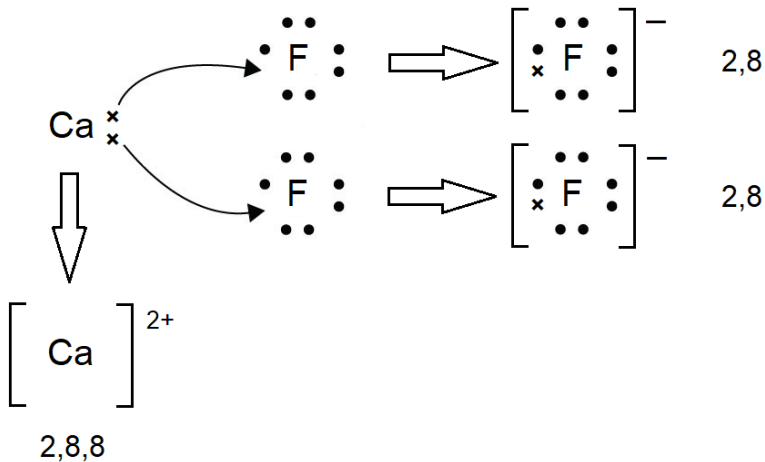


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			 award (1) for <u>transfer of electrons</u> from calcium atom to two fluorine atoms award (1) for electronic structure and charge of calcium ion award (1) for electronic structure and charge of one fluoride ion award second/third mark if both electronic structures given (no/incorrect charges) or both charges given (no/incorrect electronic structures) accept diagrams with all electron shells shown		3		3		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			award (1) for either of following - the <u>charges</u> of the <u>ions</u> in <u>calcium oxide</u> are <u>greater</u> (than the charges of the ions in potassium chloride) <u>Ca²⁺ / calcium / O²⁻ / oxide ions</u> have <u>greater charges</u> (than the K⁺ / potassium / Cl⁻ / chloride ions) [must compare size of charges] award (1) for any of following (this results in) greater attraction between the ions (this results in) stronger bonds between the ions (this means that) more energy is needed to separate the ions/break the bonds between the ions accept converse answers		2		2		
	(c)			award (1) each for any two of following - each carbon atom in diamond bonds to 4 others whereas in graphite, each carbon atom bonds to 3 others - graphite has layers but diamond doesn't / diamond has a tetrahedral structure but graphite doesn't - graphite has free electrons (between layers) whereas diamond doesn't - graphite has weak bonds between the layers whereas diamond has no weak bonds if no other credit award (1) for two partially met marking points	2			2		
				Question 4 total	2	5	0	7	0	0